

This feature could be either built into the lens itself, or into the body of the camera. There are three basic kinds of image stabilisation options in cameras – optical (built into the lens), digital (using software algorithms) or mechanical (using moving sensors).

Optical Image Stabilisation (OIS)

Cameras with optical IS use hardware to counter the movements of the camera, usually using gyro sensors and a micro-processor. In such cases, the OIS changes the optical path to the sensor when it senses vibration during exposure, and therefore, succeeds in offsetting the effect of the shake. This is built into the lens itself. The advantage here is that a camera with a lens-based VR mechanism is usually built ground up, keeping in mind the requirements of the particular lens.

Nikon and Canon cameras use a floating lens apparatus that moves orthogonally (at right angles) to the optical axis of the lens with the help of electromagnets. Shakes and jerks are perceived by gyroscopic sensors, which are basically piezoelectric angular velocity sensors (crystals that undergo mechanical stress when exposed to a fluctuating electric / electromagnetic field). This, of course, serves well for the minor vibrations of our hands – but would be no good if you rotated the camera around or jumped around with it. VR, in short, can typically only correct vertical shake.

Digital Image Stabilisation

The trick here is not to use any extra parts, either mechanical or digital at all. Digital Image Stabilisation merely heightens the camera's sensitivity by adjusting the ISO speed to get a faster shutter speed. The camera senses the vibration and does this on its own accord. The trouble – a high ISO could lead to noise and poor image quality. Rather than stabilising, digital image stabilisers just try to minimise blurring by using software akin to the ones photo editors use after the picture is taken. Digital image stabilisation only really comes of use, or makes a notable difference, in certain digital video cameras (camcorders) which 'pixel shift' the image frames to stabilise the video image while shooting; that is, buffering the motion of the camera by using pixels that are outside the visible frame of the shot.